

Module 02: Two-Phase Locking (2PL) Protocol and Lock Conversion

AKTU BCS501 | Unit 5: Concurrency Control Techniques

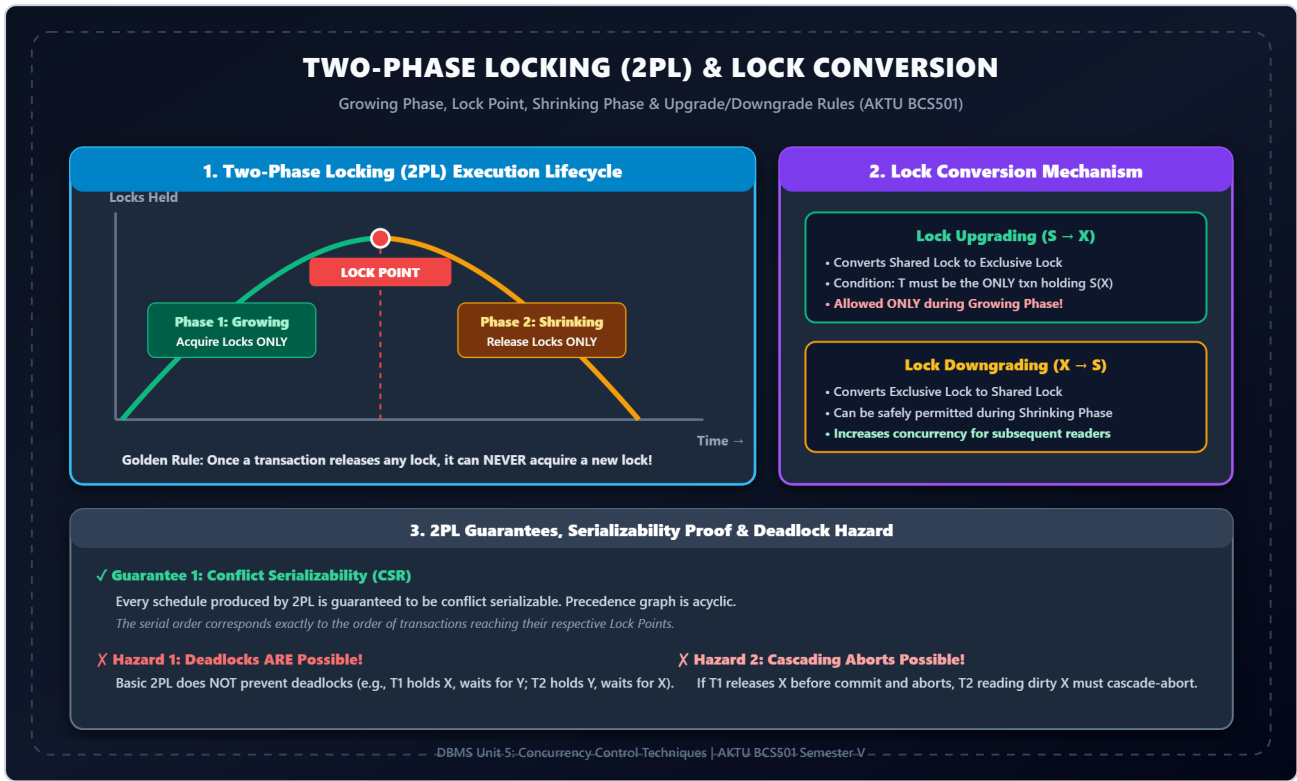


Figure 2.1: 2PL Growing Phase, Lock Point, Shrinking Phase & Conversion Rules

1. Two-Phase Locking (2PL) Definition

A transaction follows 2PL if all lock acquisitions precede the first unlock operation. It consists of a **Growing Phase** (acquire only) and a **Shrinking Phase** (release only).

Lock Point Theorem

The exact timestamp when a transaction acquires its final lock is its **Lock Point**. The equivalent serial schedule order corresponds strictly to the chronological ordering of transactions' lock points.

2. Lock Conversion (Upgrading & Downgrading)

Upgrading ($S \rightarrow X$): Allowed strictly during the Growing Phase if the transaction is the sole reader holding S(X).

Downgrading ($X \rightarrow S$): Can be safely performed in the Shrinking Phase to increase read concurrency for others.

3. 2PL Guarantees & Potential Hazards

2PL guarantees Conflict Serializability, but does NOT prevent **Deadlocks** (cyclic lock waits) or **Cascading Aborts** (if dirty locks are unlocked early).